

MUHAMMAD AYYAN

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PROFILE

Artificial Intelligence undergraduate with hands-on experience in Machine Learning, Deep Learning, Artificial Neural Networks, Natural Language Processing, and Computer Vision. Experienced in building predictive systems, AI-powered applications, and intelligent dashboards using Python and Flask. Skilled in training and evaluating Machine Learning and Deep Learning models including XGBoost, Random Forest, CNNs, ANN, Logistic Regression, Naive Bayes, and Transformer-based architectures. Proficient in data preprocessing, feature engineering, NLP techniques, and computer vision applications. Passionate about leveraging AI and data-driven technologies to solve real-world problems and enhance user experiences.

EDUCATION

COMSATS University Islamabad, Attock Campus

Bachelor of Science in Artificial Intelligence

Sep 2023 – Present

CGPA: 3.12

Relevant Courses

- Artificial Intelligence in Python
- Machine Learning in Python
- Natural Language Processing
- Computer Vision
- Artificial Neural Networks and Deep Learning
- Mobile App Development
- Information Security
- Web Technologies
- Data Structures and Algorithms
- Object-Oriented Programming

PROJECTS

AI-POWERED USER SERVICE PREDICTION SYSTEM | PYTHON, FLASK, XGBOOST, MACHINE LEARNING

Jul 2025 – Aug 2025

- Developed an AI-powered service quality monitoring system for analyzing customer service performance at Nayatel.
- Trained and optimized XGBoost and Random Forest models to predict the likelihood of future service issues.
- Applied data preprocessing, feature engineering, and exploratory data analysis on large datasets.

- Used TF-IDF vectorization and customer feedback analysis to evaluate service quality.
- Predicted the most probable fault and sub-fault categories using historical trouble ticket data.
- Built a Flask-based web dashboard with interactive UI for displaying prediction results.
- Performed testing, debugging, and model optimization for reliable real-time predictions.
- Enabled proactive customer service improvement through predictive analytics.

AI CSV ASSISTANT | PYTHON, FLASK, LANGCHAIN, GROQ LLM

Aug 2025 – Aug 2025

- Developed an AI-powered chatbot for querying CSV datasets using natural language processing.
- Built LangChain agents integrated with Groq LLM for intelligent reasoning and data interaction.
- Designed Flask backend and responsive frontend for real-time chat functionality.
- Implemented table visualization and dynamic UI components.
- Conducted debugging, testing, and optimization for improved performance.

ARABIC DIALECT DETECTION SYSTEM | PYTHON, TENSORFLOW, KERAS, NLP

March 2026 – April 2026

- Developed an AI-based Arabic Dialect Detection System to automatically identify different Arabic dialects from text data.
- Applied NLP preprocessing techniques including text cleaning, normalization, character tokenization, and feature extraction.
- Built and trained a deep learning model using TensorFlow/Keras for multi-class dialect classification.
- Implemented label encoding, model evaluation, and confusion matrix analysis to measure classification performance.
- Created a real-time prediction pipeline for detecting Arabic dialects and generating accurate classification results.

AI-BASED MOTION DETECTION SURVEILLANCE SYSTEM | PYTHON, OPENCV, YOLOV8

May 2026

- Developed a real-time surveillance system using OpenCV and YOLOv8.
- Implemented image preprocessing, contour analysis, and motion detection techniques.
- Built an interactive desktop GUI using Tkinter.
- Enabled automatic image capture and object localization.

SMART STUDENT PLANNER MOBILE APP | REACT NATIVE, EXPO

May 2026

- Developed a cross-platform mobile application using React Native and Expo.
 - Implemented Stack, Tab, and Drawer Navigation.
 - Designed interactive dashboards for schedules, assignments, and task management.
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WORK EXPERIENCE

Artificial Intelligence Intern

Nayatel Pvt. Ltd., Islamabad, Pakistan

Jul 2025 – Aug 2025

Completed a 6-week internship at Nayatel Pvt. Ltd., working on AI-driven customer service analytics and predictive systems.

- Designed and trained XGBoost and Random Forest models.
- Applied NLP techniques and TF-IDF vectorization on customer feedback.
- Performed data preprocessing, visualization, and feature engineering.
- Developed Flask APIs and dashboard integration.
- Conducted model testing, debugging, and optimization.
- Gained practical exposure to Machine Learning, Deep Learning, NLP, Transformers, and AI workflows.

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, JavaScript
- **Machine Learning:** Scikit-Learn, XGBoost, Random Forest, Logistic Regression, Naive Bayes, Support Vector Machine, KNN
- **Deep Learning:** Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), TensorFlow, Keras, PyTorch
- **Natural Language Processing:** TF-IDF, Text Classification, Tokenization, Named Entity Recognition (NER), Sentence-BERT, Hugging Face Transformers, KoBERT
- **LLMs & AI Frameworks:** LangChain, LangGraph, Groq LLM, Semantic Search, AI Chatbots
- **Computer Vision:** OpenCV, YOLOv8, Object Detection, Motion Detection, Image Processing
- **Data Science & Analytics:** Pandas, NumPy, Matplotlib, Feature Engineering, Data Preprocessing, Exploratory Data Analysis, Model Evaluation
- **Web Development:** Flask, HTML, CSS, Tailwind CSS, REST APIs, Dashboard Development, Responsive UI Design
- **Mobile App Development:** React Native, Expo, React Hooks, State Management, Stack Navigation, Tab Navigation, Drawer Navigation
- **Tools & Platforms:** Git, GitHub, Jupyter Notebook, VS Code, Joblib, Virtual Environments
- **Testing & Optimization:** Debugging, Performance Optimization, Model Integration, Real-Time Prediction Systems
- **Additional Skills:** CSV Data Handling, PDF Processing, Multilingual Translation Systems, Technical Documentation

PROFESSIONAL SKILLS

- Problem Solving and Analytical Thinking
- Team Collaboration and Communication
- Technical Documentation and Reporting
- Presentation and Interpersonal Skills
- Adaptability and Continuous Learning
- Time Management and Leadership

CERTIFICATE

- **Artificial Intelligence Internship** – Nayatel Pvt. Ltd. (2025)